

Online Retail Dataset Analysis – One-Page Summary Report

Data Cleaning

Several data quality issues were identified and addressed before analysis. The CustomerID column contained a significant number of missing values, while a small number of records had missing product descriptions. Duplicate transactions were also present and could potentially distort analytical results. In addition, some records contained invalid values, such as negative or zero quantities and unit prices, which mainly represented returns, cancellations, or data-entry errors. Text inconsistencies, including variations in capitalization and extra spaces within the Description and Country fields, were also observed. Furthermore, invoice numbers beginning with the letter “C” indicated canceled transactions.

To improve data quality, missing and duplicate records were removed, invalid entries were filtered out, and text and date formats were standardized. These cleaning steps ensured that the dataset was accurate, consistent, and suitable for reliable analysis.

Exploratory Data Analysis (EDA) Findings

The dataset contains over 500,000 retail transactions involving thousands of products and customers across multiple countries. Analysis revealed that sales are highly concentrated, with a relatively small number of products and customers generating a large proportion of transactions and revenue.

The United Kingdom dominates the dataset, contributing the majority of sales activity. Customer purchasing behavior is strongly right-skewed, meaning most customers make only a few purchases, while a small group of customers place orders frequently and contribute significantly to overall revenue.

Monthly sales trend analysis showed clear seasonality, with sales increasing steadily throughout the year and reaching their highest levels between August and November. This pattern suggests the influence of holiday shopping and seasonal demand.

Key Insights and Conclusion

A small number of products account for a large share of total sales, with items such as World War 2 Gliders and Jumbo Bag Red Retrospot among the top-selling products. Customer activity is similarly concentrated, highlighting the importance of retaining loyal and high-value customers. Sales trends indicate strong seasonal demand, peaking in November before declining in December, which may be partially due to incomplete data.

Overall, the analysis demonstrates that business performance is driven by a combination of popular products, repeat customers, and seasonal purchasing patterns. Effective data cleaning was essential for uncovering these insights, and the cleaned dataset provides a reliable foundation for business decision-making and future analytical work.